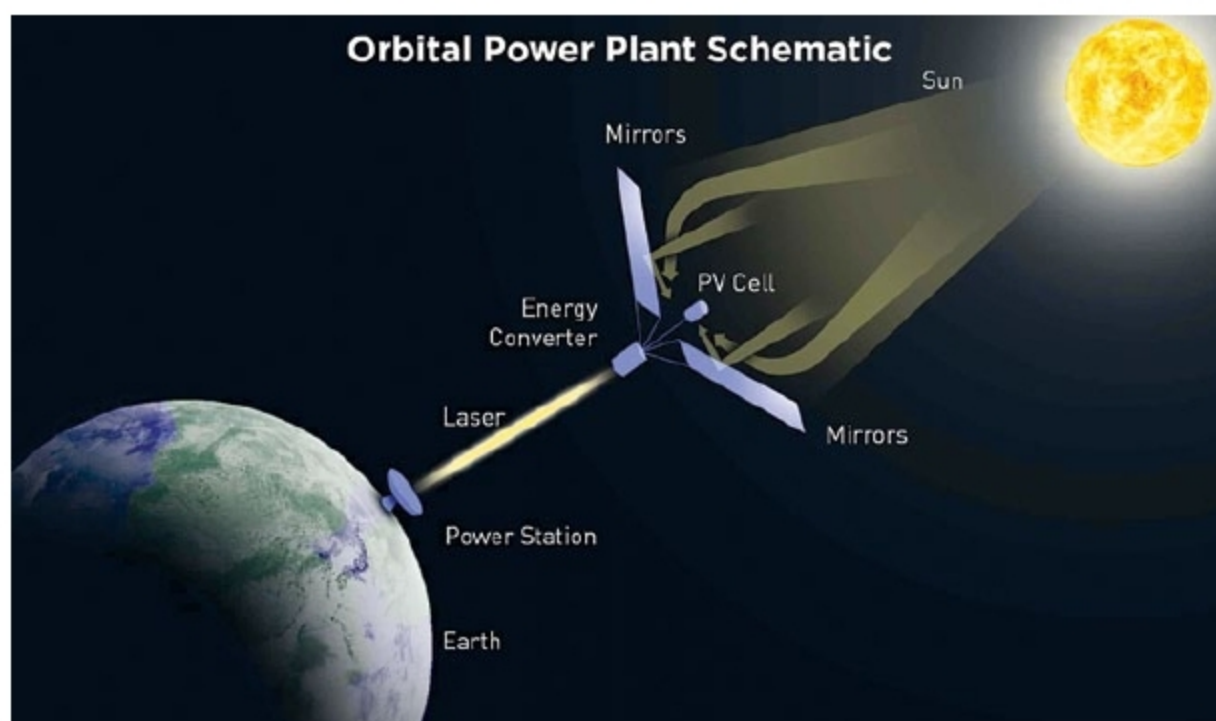




- 1 Resurgence Of Space-Based Solar Power
- 2 Bio-CNG: The Panacea For Green Transportation?
- 3 Becoming Financially Smarter With New-Age Banking
- 4 Parking 4.0: The Next-Generation Parking Management System

1 RESURGENCE OF SPACE-BASED SOLAR POWER

If successful, space-based solar systems can be sustainable and inexhaustible long-term solutions for meeting power requirements for the entire world



(Credit: <https://futurism.com>)

Solar panel arrays have been used to power space stations and satellites for a long time now, but the concept of harnessing solar power from outer space is still at a nascent stage. On Earth, solar arrays are used for power generation during daytime in places with enough sunlight. Many places like Russia do not receive enough sunlight all year round. Besides, a huge portion of solar energy reaching the planet is lost due to reflection and absorption by dust and other particles in the atmosphere.

Solar energy has to be stored in batteries at present as a backup for harsh weather and night-time. If suc-

cessful, space-based solar systems can be sustainable and inexhaustible long-term solutions for meeting power requirements for the entire world.

Compared to terrestrial solar farms, these systems can receive permanent sunlight 24/7 and capture sun rays without any hindrance, providing huge output capacity. This power generated at a continuous pace can then be instantly transmitted anywhere across the world, such as in remote areas for military, decreasing the dependence on batteries.

This fascinating idea was attempted in the 1970s, but several barriers made it infeasible to pursue it further. Even

today, the most daunting challenge is the cost incurred with no assurance of returns. Solar-powered satellites require a large array of solar panels, which also means higher mass.

Materials used in manufacturing have a short lifetime in space due to stringent conditions, and large amounts of materials requiring quick replacement is a big hassle. The solution is to make them lightweight without decreasing efficiency.

Despite the challenges, Indian Space Research Organisation (ISRO) has partnered with US National Space Society and with China for development of solar collectors and solar satellites. In 2015, Japanese Aerospace Exploration Agency (JAXA) successfully transmitted 1.8 kilowatts of energy using microwaves across a distance of 55 metres to a receiving antenna. The agency has been researching on overcoming technological challenges like those associated with wireless power transmissions.

In February 2019, China Association for Science and Technology (CAST) announced its plans of building power satellites. The vision of various countries is to build a commercial-scale solar power station in space in the next few decades.

Many designs for capture and transmission of solar energy have been proposed by researchers. Most of them focus on utilising optical